# MSG No. 196.4 $F23$ [ Type I, cubic ]

* symmetry operation

Table 1: Symmetry operations for 3d polar vector.

| No. | tag | matrix (polar) | det | TR |
|---|---|---|---|---|
| 1 | $\{1\|0\}$ | $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$ | 1 | 1 |
| 2 | $\{2_{100}\|0\}$ | $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$ | 1 | 1 |
| 3 | $\{2_{010}\|0\}$ | $\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$ | 1 | 1 |
| 4 | $\{2_{001}\|0\}$ | $\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$ | 1 | 1 |
| 5 | $\{3^{+}_{111}\|0\}$ | $\begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$ | 1 | 1 |
| 6 | $\{3^{-}_{111}\|0\}$ | $\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$ | 1 | 1 |
| 7 | $\{3^{-}_{1-1-1}\|0\}$ | $\begin{bmatrix} 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 0 \end{bmatrix}$ | 1 | 1 |
| 8 | $\{3^{+}_{1-1-1}\|0\}$ | $\begin{bmatrix} 0 & 0 & -1 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$ | 1 | 1 |
| 9 | $\{3^{-}_{-11-1}\|0\}$ | $\begin{bmatrix} 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$ | 1 | 1 |
| 10 | $\{3^{+}_{-11-1}\|0\}$ | $\begin{bmatrix} 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix}$ | 1 | 1 |
| 11 | $\{3^{-}_{-1-11}\|0\}$ | $\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ -1 & 0 & 0 & 0 \end{bmatrix}$ | 1 | 1 |
| 12 | $\{3^{+}_{-1-11}\|0\}$ | $\begin{bmatrix} 0 & 0 & -1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix}$ | 1 | 1 |
| 13 | $\{1\|0\frac{1}{2}\frac{1}{2}\}$ | $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$ | 1 | 1 |

*continued ...*

Table 1

| No. | tag | matrix (polar) | det | TR |
|---|---|---|---|---|
| 14 | $\{2_{100}\|0\frac{1}{2}\frac{1}{2}\}$ | $\begin{bmatrix}1 & 0 & 0 & 0\\ 0 & -1 & 0 & \frac{1}{2}\\ 0 & 0 & -1 & \frac{1}{2}\end{bmatrix}$ | 1 | 1 |
| 15 | $\{2_{010}\|0\frac{1}{2}\frac{1}{2}\}$ | $\begin{bmatrix}-1 & 0 & 0 & 0\\ 0 & 1 & 0 & \frac{1}{2}\\ 0 & 0 & -1 & \frac{1}{2}\end{bmatrix}$ | 1 | 1 |
| 16 | $\{2_{001}\|0\frac{1}{2}\frac{1}{2}\}$ | $\begin{bmatrix}-1 & 0 & 0 & 0\\ 0 & -1 & 0 & \frac{1}{2}\\ 0 & 0 & 1 & \frac{1}{2}\end{bmatrix}$ | 1 | 1 |
| 17 | $\{3^{+}_{111}\|0\frac{1}{2}\frac{1}{2}\}$ | $\begin{bmatrix}0 & 0 & 1 & 0\\ 1 & 0 & 0 & \frac{1}{2}\\ 0 & 1 & 0 & \frac{1}{2}\end{bmatrix}$ | 1 | 1 |
| 18 | $\{3^{-}_{111}\|0\frac{1}{2}\frac{1}{2}\}$ | $\begin{bmatrix}0 & 1 & 0 & 0\\ 0 & 0 & 1 & \frac{1}{2}\\ 1 & 0 & 0 & \frac{1}{2}\end{bmatrix}$ | 1 | 1 |
| 19 | $\{3^{-}_{1-1-1}\|0\frac{1}{2}\frac{1}{2}\}$ | $\begin{bmatrix}0 & -1 & 0 & 0\\ 0 & 0 & 1 & \frac{1}{2}\\ -1 & 0 & 0 & \frac{1}{2}\end{bmatrix}$ | 1 | 1 |
| 20 | $\{3^{+}_{1-1-1}\|0\frac{1}{2}\frac{1}{2}\}$ | $\begin{bmatrix}0 & 0 & -1 & 0\\ -1 & 0 & 0 & \frac{1}{2}\\ 0 & 1 & 0 & \frac{1}{2}\end{bmatrix}$ | 1 | 1 |
| 21 | $\{3^{-}_{-11-1}\|0\frac{1}{2}\frac{1}{2}\}$ | $\begin{bmatrix}0 & -1 & 0 & 0\\ 0 & 0 & -1 & \frac{1}{2}\\ 1 & 0 & 0 & \frac{1}{2}\end{bmatrix}$ | 1 | 1 |
| 22 | $\{3^{+}_{-11-1}\|0\frac{1}{2}\frac{1}{2}\}$ | $\begin{bmatrix}0 & 0 & 1 & 0\\ -1 & 0 & 0 & \frac{1}{2}\\ 0 & -1 & 0 & \frac{1}{2}\end{bmatrix}$ | 1 | 1 |
| 23 | $\{3^{-}_{-1-11}\|0\frac{1}{2}\frac{1}{2}\}$ | $\begin{bmatrix}0 & 1 & 0 & 0\\ 0 & 0 & -1 & \frac{1}{2}\\ -1 & 0 & 0 & \frac{1}{2}\end{bmatrix}$ | 1 | 1 |
| 24 | $\{3^{+}_{-1-11}\|0\frac{1}{2}\frac{1}{2}\}$ | $\begin{bmatrix}0 & 0 & -1 & 0\\ 1 & 0 & 0 & \frac{1}{2}\\ 0 & -1 & 0 & \frac{1}{2}\end{bmatrix}$ | 1 | 1 |
| 25 | $\{1\|\frac{1}{2}0\frac{1}{2}\}$ | $\begin{bmatrix}1 & 0 & 0 & \frac{1}{2}\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & \frac{1}{2}\end{bmatrix}$ | 1 | 1 |
| 26 | $\{2_{100}\|\frac{1}{2}0\frac{1}{2}\}$ | $\begin{bmatrix}1 & 0 & 0 & \frac{1}{2}\\ 0 & -1 & 0 & 0\\ 0 & 0 & -1 & \frac{1}{2}\end{bmatrix}$ | 1 | 1 |
| 27 | $\{2_{010}\|\frac{1}{2}0\frac{1}{2}\}$ | $\begin{bmatrix}-1 & 0 & 0 & \frac{1}{2}\\ 0 & 1 & 0 & 0\\ 0 & 0 & -1 & \frac{1}{2}\end{bmatrix}$ | 1 | 1 |

*continued ...*

Table 1

| No. | tag | matrix (polar) | det | TR |
|---|---|---|---|---|
| 28 | $\{2_{001}\|\frac{1}{2}0\frac{1}{2}\}$ | $\begin{bmatrix} -1 & 0 & 0 & \frac{1}{2} \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$ | 1 | 1 |
| 29 | $\{3^{+}_{111}\|\frac{1}{2}0\frac{1}{2}\}$ | $\begin{bmatrix} 0 & 0 & 1 & \frac{1}{2} \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & \frac{1}{2} \end{bmatrix}$ | 1 | 1 |
| 30 | $\{3^{-}_{111}\|\frac{1}{2}0\frac{1}{2}\}$ | $\begin{bmatrix} 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & \frac{1}{2} \end{bmatrix}$ | 1 | 1 |
| 31 | $\{3^{-}_{1-1-1}\|\frac{1}{2}0\frac{1}{2}\}$ | $\begin{bmatrix} 0 & -1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & \frac{1}{2} \end{bmatrix}$ | 1 | 1 |
| 32 | $\{3^{+}_{1-1-1}\|\frac{1}{2}0\frac{1}{2}\}$ | $\begin{bmatrix} 0 & 0 & -1 & \frac{1}{2} \\ -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & \frac{1}{2} \end{bmatrix}$ | 1 | 1 |
| 33 | $\{3^{-}_{-11-1}\|\frac{1}{2}0\frac{1}{2}\}$ | $\begin{bmatrix} 0 & -1 & 0 & \frac{1}{2} \\ 0 & 0 & -1 & 0 \\ 1 & 0 & 0 & \frac{1}{2} \end{bmatrix}$ | 1 | 1 |
| 34 | $\{3^{+}_{-11-1}\|\frac{1}{2}0\frac{1}{2}\}$ | $\begin{bmatrix} 0 & 0 & 1 & \frac{1}{2} \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & \frac{1}{2} \end{bmatrix}$ | 1 | 1 |
| 35 | $\{3^{-}_{-1-11}\|\frac{1}{2}0\frac{1}{2}\}$ | $\begin{bmatrix} 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & -1 & 0 \\ -1 & 0 & 0 & \frac{1}{2} \end{bmatrix}$ | 1 | 1 |
| 36 | $\{3^{+}_{-1-11}\|\frac{1}{2}0\frac{1}{2}\}$ | $\begin{bmatrix} 0 & 0 & -1 & \frac{1}{2} \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & \frac{1}{2} \end{bmatrix}$ | 1 | 1 |
| 37 | $\{1\|\frac{1}{2}\frac{1}{2}0\}$ | $\begin{bmatrix} 1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 \end{bmatrix}$ | 1 | 1 |
| 38 | $\{2_{100}\|\frac{1}{2}\frac{1}{2}0\}$ | $\begin{bmatrix} 1 & 0 & 0 & \frac{1}{2} \\ 0 & -1 & 0 & \frac{1}{2} \\ 0 & 0 & -1 & 0 \end{bmatrix}$ | 1 | 1 |
| 39 | $\{2_{010}\|\frac{1}{2}\frac{1}{2}0\}$ | $\begin{bmatrix} -1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & -1 & 0 \end{bmatrix}$ | 1 | 1 |
| 40 | $\{2_{001}\|\frac{1}{2}\frac{1}{2}0\}$ | $\begin{bmatrix} -1 & 0 & 0 & \frac{1}{2} \\ 0 & -1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 \end{bmatrix}$ | 1 | 1 |
| 41 | $\{3^{+}_{111}\|\frac{1}{2}\frac{1}{2}0\}$ | $\begin{bmatrix} 0 & 0 & 1 & \frac{1}{2} \\ 1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & 0 \end{bmatrix}$ | 1 | 1 |

*continued ...*

Table 1

| No. | tag | matrix (polar) | det | TR |
|---|---|---|---|---|
| 42 | $\{3^{-}_{111}\vert\frac{1}{2}\frac{1}{2}0\}$ | $\begin{bmatrix} 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} \\ 1 & 0 & 0 & 0 \end{bmatrix}$ | 1 | 1 |
| 43 | $\{3^{-}_{1-1-1}\vert\frac{1}{2}\frac{1}{2}0\}$ | $\begin{bmatrix} 0 & -1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} \\ -1 & 0 & 0 & 0 \end{bmatrix}$ | 1 | 1 |
| 44 | $\{3^{+}_{1-1-1}\vert\frac{1}{2}\frac{1}{2}0\}$ | $\begin{bmatrix} 0 & 0 & -1 & \frac{1}{2} \\ -1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & 0 \end{bmatrix}$ | 1 | 1 |
| 45 | $\{3^{-}_{-11-1}\vert\frac{1}{2}\frac{1}{2}0\}$ | $\begin{bmatrix} 0 & -1 & 0 & \frac{1}{2} \\ 0 & 0 & -1 & \frac{1}{2} \\ 1 & 0 & 0 & 0 \end{bmatrix}$ | 1 | 1 |
| 46 | $\{3^{+}_{-11-1}\vert\frac{1}{2}\frac{1}{2}0\}$ | $\begin{bmatrix} 0 & 0 & 1 & \frac{1}{2} \\ -1 & 0 & 0 & \frac{1}{2} \\ 0 & -1 & 0 & 0 \end{bmatrix}$ | 1 | 1 |
| 47 | $\{3^{-}_{-1-11}\vert\frac{1}{2}\frac{1}{2}0\}$ | $\begin{bmatrix} 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & -1 & \frac{1}{2} \\ -1 & 0 & 0 & 0 \end{bmatrix}$ | 1 | 1 |
| 48 | $\{3^{+}_{-1-11}\vert\frac{1}{2}\frac{1}{2}0\}$ | $\begin{bmatrix} 0 & 0 & -1 & \frac{1}{2} \\ 1 & 0 & 0 & \frac{1}{2} \\ 0 & -1 & 0 & 0 \end{bmatrix}$ | 1 | 1 |